THE ODYNEX INSTITUTE AT FORT GAINES

A Strategic Blueprint for Regional Data-Resiliency and Ecosystem Co-Development
In Sister Partnership with Columbus State University

1. Executive Cover Letter

Date: August 16, 2026
From: The Planning Department, City of Fort Gaines, GA
To: Office of the President, Columbus State University
Subject: Proposal for Academic Partnership: The Odynex Institute Rural Field Laboratory

Dear President Rayfield,

Please find attached the proposed Memorandum of Understanding (MOU) establishing a formal Sister Institution partnership between Columbus State University and the City of Fort Gaines, Georgia.

With the recent state commitment toward your **$22.2 Million Center for Advanced Technologies**, Columbus State University is positioned to lead the nation in applied robotics and computational engineering. To maximize the scope of this new facility, our proposed rural outpost—the Odynex Institute—offers CSU an open-air field laboratory to test automated resource telemetry, micro-grids, and amphibious architectures in a live environment.

Furthermore, as the region navigates community questions surrounding the **$5 Billion Project Ruby Data Center Campus**, this field lab provides an active testbed for non-mechanical water filtration and sustainable cooling templates. This directly equips regional developers with empirical data to address infrastructure and resource management concerns.

Please find enclosed the formal structural framework, curriculum configurations, and first-year budgetary allocations. We look forward to collaborating to build an innovative workforce pipeline for the Chattahoochee Valley.

Sincerely,

The Fort Gaines Adaptive Bioeconomy Initiative Coordinating Committee

2. Memorandum of Understanding (MOU) Template

The following contractual layout details the joint governance framework between the partner organizations:

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MEMORANDUM OF UNDERSTANDING (MOU)
BETWEEN COLUMBUS STATE UNIVERSITY & THE CITY OF FORT GAINES, GEORGIA
REGARDING THE ESTABLISHMENT OF THE ODYNEX INSTITUTE RURAL FIELD LAB

1. PURPOSE & OBJECTIVE
This agreement formally establishes an academic, scientific, and operational
partnership
between Columbus State University (hereafter 'CSU') and the City of Fort Gaines, GA
(hereafter 'Fort Gaines') to develop the Odynex Institute. This facility will serve
as
CSU's primary Rural Field Laboratory for amphibious architecture, micro-grid
telemetry,
and circular bioeconomy systems.

2. COOPERATIVE FRAMEWORK & RESPONSIBILITIES
A. Columbus State University agrees to:
i. Provide academic oversight and coordinate undergraduate/graduate student
placement.
ii. Integrate telemetry streams from the Fort Gaines Pod cluster into the CSU
Advanced
Technologies, Robotics, and Space Science Facility Mainframe.
iii. Jointly apply for University System of Georgia (USG) and state-level
multidisciplinary
innovation grants.

B. The City of Fort Gaines / Odynex Axis Logistics agrees to:
i. Dedicate underutilized public right-of-ways and flood-prone municipal bluff zones
for the campus layout.
ii. Provide structural maintenance, safety, and local emergency management
coordination
for the amphibious pod clusters and interactive family eco-trails.
iii. Facilitate direct partnerships with local agricultural stakeholders for biomass
collection.

3. RESOURCE & FACILITY INTEGRATION
Both parties agree that the Fort Gaines campus will operate on a 'visible-systems'
architecture. Students in residence will actively perform structural, environmental,
and
computational upgrades to the facility as a core requirement of experiential
learning.

4. DURATION AND AMENDMENT
This MOU shall remain in effect for a period of five (5) years from the date of final
signature,
subject to annual performance reviews and mutual expansion updates.
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3. Joint Academic Curriculum Outline

Course Title: ENGR / BIOL 4890: Experiential Rural Infrastructure & Biomass Dynamics
Prerequisites: Completion of lower-division technical/science core at CSU Main Campus.
Residency Location: Odynex Institute Pod Cluster, Fort Gaines, GA.

• Weeks 1–4: Telemetry Calibration & Micro-Grid Baselines

Students audit their assigned living pod clusters. They map color-coded low-voltage DC lines and mount autonomous IoT micro-controllers. *Measurable Output:* A running telemetry panel checking energy input versus load metrics.

• Weeks 5–10: Bio-Sluice Optimization & Biomass Scaling

Students construct and test physical sand, charcoal, and clay matrix filter variants within the living-lab greywater lines. Simultaneously, they feed peanut hulls, agricultural waste, and yard clippings into insect farming enclosures. *Measurable Output:* A laboratory portfolio checking fluid purity metrics (pH, TDS) and larval organic breakdown speeds.

• Weeks 11–15: Amphibious Stress Testing & Public Reporting

Students assess the buoyancy mechanics and physical clearance of the float-track guide blocks. *Measurable Output:* Final structural presentations to regional technical bodies and the Fort Gaines City Council.

4. Technical Specification & Data Node Guide

To establish cross-county operational continuity, each modular cell functions as an edge node transmitting continuous metrics to the mainframe in Columbus:

A. Micro-Grid & Power Telemetry: Adafruit INA219 high-side current sensors are integrated inline with the low-voltage lines to process panel input watt-hours and localized battery cell status.

B. Fluid & Bio-Filtration Monitoring: Industrial analog pH meters and waterproof DS18B20 temperature probes evaluate bio-sluice performance data at a 1Hz sampling interval.

C. Network Architecture: Embedded ESP32 micro-controllers handle edge computational conversions, pushing metrics through a Dragino LPS8v2 Outdoor LoRaWAN Gateway via encrypted MQTT brokers to AWS IoT Core Mainframes.

5. First-Year Pilot Operating Budget Sheet

The initial phase operates on a cost-sharing architecture leveraging state-level safety, infrastructure, and academic allocations:

Category	Line-Item Description	First-Year CapEx	Annual OpEx	Funding Channel Hook
Telemetry Hardware	IoT sensor boards, flow meters, LoRa gateways	$4,500	$600	USG Equipment Grant

Amphibious Tracks	Concrete pontoons & vertical structural guide beams	$18,500	$1,200	GEMA Flood Mitigation
Bio-Aqua Infra	Permeable volcanic soils, stone sluices, fingerlings	$6,200	$1,800	Agricultural Seed Capital
Student Ops	Local utilities, emergency satellite backup link	$0 (Exempt)	$3,400	University Housing Funds
Workforce Lab Supplies	Test kits, safety gear, insect breeding stocks	$2,100	$1,500	Corporate Manufacturing
TOTALS	Combined Pilot System Costs	$31,300	$8,500	Joint Regional Capital

This is for informational and strategic planning purposes only. AI responses may include mistakes.